ONLINE PAYMENT FRAUD DETECTION USING MACHING LEARNING TECHNIQUES

A Project Report

Submitted

In partial fulfillment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE and ENGINEERING

By

Sk.Adil Azeez	(201FA04331)
P.Veda Phani	(211LA04012)
G.Durga Praveen	(211LA04002)

Under the guidance of

Dr.Saubhagya Ranjan Biswal

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

VIGNAN'S FOUNDATION FOR SCIENCE, TECHNOLOGY AND RESEARCH

Deemed to be UNIVERSITY Vadlamudi, Guntur.

ANDHRA PRADESH, INDIA, PIN-522 213

MAY-2024.

VIGNAN'S
Foundation for Science, Technology & Research
(Deemed to be University)
-Estd. u/s 3 of UGC Act 1956

CERTIFICATE

This is to certify that the report entitled **"Online Payment Fraud Detection Using Machine Learning Techniques"** is submitted by **"Sk.Adil Azeez (201FA04331),P.Veda Phani (211LA04012), G.Durga Praveen (211LA04002)"** in the partial fulfillment of course work of a major project carried out in the department of CSE,VFSTR Deemed to be University.

Dr.Saubhagya Ranjan Biswal	**External Examiner**	**Dr.K. V. Krishna Kishore**
Assistant Professor		HOD, CSE
		Dean,School of Comp,& Info

DECLARATION

We hereby declare that the project entitled "**Online Payment Fraud Detection Using Machine Learning Techniques**" submitted to the "**Department of Computer Science and Engineering"**. This dissertation is our original work and the project has not formed the basis for the award of any degree, associate-ship and fellowship or any other similar titles and no part of it has been published or sent for publication at the time of submission.

By

Sk. Adil Azeez (201FA04331),

P. Veda Phani (211LA04012),

G. Durga Praveen (211LA04002).

Signature(s):

Date:

ACKNOWLEDGEMENT

This project program is a golden opportunity for learning and self-development. Consider ourselves very lucky and honored to have so many wonderful people lead us through in the completion of this project.

We extend our sincere appreciation to the Project Guide **Dr. Saubhagya Ranjan Biswal**, for granting us permission to undertake the Program and for his continuous support and cooperation throughout the duration of our project.

We express our gratitude to the Project Coordinator **Dr. Yakobu Dasari**, for permitting us to undertake the Program and for the help and cooperation throughout the course of our project Program.

It is a great pleasure for us to express our sincere thanks to **Dr. K. V. Krishna Kishore, HOD, CSE and Dean, School of Computing & Informatics** of VFSTR Deemed to be a university, for providing us with an opportunity to do our Project Program.

We extend our wholehearted gratitude to all our faculty members, programmers, and technicians of the Department of Computer Science and Engineering who helped using our academics throughout the course.

Finally, we wish to express thanks to our family members for the love and affection overseas and cheerful depositions, which are vital for sustaining the effort required for completing this work.

Sk.Adil Azeez (201FA04331)
P.Veda Phani (211LA04012)
G.Durga Praveen (211LA04002)

ABSTRACT

With the exponential growth of online transactions, the risk of fraudulent activities has become a significant concern for both consumers and businesses. Traditional rule-based systems for fraud detection are often inadequate in identifying sophisticated fraudulent patterns, leading to substantial financial losses. In this paper, we propose an enhanced online payment fraud detection system leveraging advanced machine learning techniques. Our approach involves the utilization of various machine learning algorithms to analyze transactional data and detect anomalous patterns indicative of fraudulent behavior. Feature engineering techniques are employed to extract relevant information from transactional data, while ensemble learning methods are utilized to improve the robustness and accuracy of the detection system**.** Furthermore, the incorporation of anomaly detection algorithms to identify previously unseen fraudulent activities is also explored. The effectiveness of the proposed approach is evaluated using real-world transactional datasets, demonstrating superior performance compared to traditional rule-based systems. Through extensive experimentation and analysis, the potential of machine learning techniques in bolstering the security of online payment systems is illustrated by efficiently detecting fraudulent activities while minimizing false positives.

Keywords: Payment Fraud Detection, Machine Learning, Decision Tree, Logistic Regression, Support Vector Machines, Random Forest, Accuracy

TABLE OF CONTENTS

1. Introduction 1
   1.1 Motivation 1
   1.2 Problem Definition 1
   1.3 Constraints 2
   1.4 Design Standards 3
   1.5 Objectives 3
   1.6 Overall Organization of thesis 4
2. Related Work 5
   2.1 Recent Literature 6
   2.2 Limitations 10
3. Proposed System 11
   3.1 Proposed Workflow 13
   3.2 Discussion of Proposed workflow Diagram 13
   3.3 Data Analysis 14
      3.3.1 Analyzing Missing Values 14
      3.3.2 Target Distribution Investigation 15
      3.3.3 Relation Between Characteristics 15
      3.3.4 Data Preparation 16
      3.3.5 Feature Selection 16
      3.3.6 Handling Class Imbalance 16
   3.4 Modelling Approach 18
      3.4.1 Supervised Learning 18
      3.4.2 Support Vector Machine 20
      3.4.3 Logistic Regression 21
      3.4.4 Random Forest 22
      3.4.5 Decision Tree 23
4. Experiment Results and Discussions 24
   4.1 Dataset Details 24
   4.2 Performance Evaluation 24
   4.3 Discussions 26
      4.3.1 Hyper Parameters 26
      4.3.2 Plots and Confusion Matrix 27
      4.3.3 Comparative Analysis 28
      4.3.4 Sample Output 29
5. Conclusion and Future Scope 31
   5.1 Conclusion 31
   5.2 Future Scope 31
   References 32

LIST OF FIGURES

Figure 3.1 System Workflow 11

Figure 3.2 Architecture 12

Figure 3.3 Dataset 13

Figure 3.4 Missing Values 14

Figure 3.5 Imbalance Data 14

Figure 3.6 Correlation 15

Figure 3.7 Target Feature 17

Figure 3.8 Machine Learning Techniques 17

Figure 3.9 Support Vector Machines 19

Figure 3.10 Logistic Regression 20

Figure 3.11 Random Forest 21

Figure 3.12 Decision Tree 22

Figure 4.1 Confusion Matrix 24

Figure 4.2 Transaction Amount 27

Figure 4.3 Transaction step 27

Figure 4.4 Old Balance 27

Figure 4.5 New Balance 27

Figure 4.6 Confusion Matrix on Proposed Model. 28

Figure 4.7 Prediction Output 30

LIST OF TABLES

Table 2.1: Summary of Existing Approaches 10

Table 4.1: Hyperparameters 26

Table 4.2: Comparative Analysis 29

CHAPTER-1
INTRODUCTION

1.1 Motivation

The proliferation of online transactions has revolutionized the way we conduct business, enabling seamless transactions across geographical boundaries and enhancing convenience for consumers worldwide. However, this digital revolution has also brought about new challenges, particularly concerning the security of online payment systems. One of the most pressing concerns is the prevalence of fraudulent activities, which can result in substantial financial losses for both consumers and businesses alike.

Traditional methods of fraud detection, often reliant on predefined rules and thresholds, are increasingly ineffective in combating the evolving tactics employed by fraudsters. These rule-based systems struggle to adapt to the dynamic nature of fraudulent behavior, leading to a high rate of false positives and missed detections. As a result, there is a growing need for more advanced and adaptable approaches to online payment fraud detection.

In recent years, machine learning has emerged as a powerful tool in the fight against fraud, offering the ability to analyze large volumes of transactional data and identify complex patterns indicative of fraudulent behavior. By leveraging the vast amounts of data generated by online transactions, machine learning algorithms can detect subtle anomalies and deviations from normal behavior, thereby flagging potentially fraudulent activities in real-time.

1.2 Problem Definition

Fraudsters employ a myriad of sophisticated techniques to perpetrate online payment fraud. These methods include phishing attacks, account takeovers, and the use of synthetic identities. Each technique has unique characteristics and evolves over time, making it increasingly difficult to detect using traditional methods. The dynamic nature of these fraudulent activities requires adaptive and intelligent detection systems capable of staying ahead of emerging threats. Additionally, the sheer volume and speed of online transactions necessitate a real-time detection approach that can operate at scale, efficiently processing millions of transactions daily to identify

potentially fraudulent activities.

One of the significant challenges in online payment fraud detection is the issue of data imbalance. Fraudulent transactions constitute a small fraction of the total transaction volume, creating a highly imbalanced dataset. This imbalance can lead to machine learning models being biased towards predicting non-fraudulent transactions, thereby reducing the overall effectiveness of the fraud detection system. Moreover, traditional rule-based systems often fall short as they rely on predefined patterns and heuristics, which can be too rigid to capture the nuances of evolving fraud tactics. These systems also tend to generate high false positive rates, which can inconvenience legitimate users and degrade their trust in the payment system.

1.3 Constraints

Data Quality:

The effectiveness of machine learning models heavily relies on the quality of the data used for training.

Poor quality data with errors, missing values, or inaccuracies can lead to unreliable models. Ensuring clean, consistent, and high-quality data is a major challenge.

Data Availability:

Comprehensive datasets that include diverse examples of both legitimate and fraudulent transactions are essential for training effective models. However, collecting sufficient fraudulent transaction data is difficult due to the inherent rarity of such events, leading to data imbalance issues.

Privacy Concerns:

Handling sensitive financial data requires strict adherence to privacy regulations such as GDPR, CCPA, and other data protection laws. Ensuring that data is anonymized and securely stored while being used for training machine learning models is crucial.

Security Risks:

The data used for fraud detection must be protected against unauthorized access and breaches. Ensuring data security throughout the lifecycle of the machine learning model, from data collection to deployment, is essential to maintain trust and compliance.

Latency and Speed:

Online payment systems require real-time or near-real-time fraud detection to prevent

fraudulent transactions from being processed. Machine learning models must be optimized to deliver quick and accurate decisions without introducing significant delays in transaction processing.

Scalability:

The system must handle large volumes of transactions simultaneously, especially during peak times such as holidays or sales events. Scalability of the machine learning model and the underlying infrastructure is critical to ensure consistent performance.

System Compatibility:

Integrating machine learning models with existing payment processing systems and fraud detection frameworks can be challenging. Ensuring compatibility and seamless integration is necessary to avoid disruptions and maintain operational efficiency.

Interoperability:

The fraud detection system must be able to work across various platforms, payment gateways, and financial institutions, requiring a high degree of interoperability and standardization.

Model Maintenance:

Fraud patterns evolve over time, necessitating continuous updates and retraining of machine learning models. Maintaining and updating models to adapt to new fraud techniques requires ongoing resources and expertise.

1.4 Design Standards

Algorithm Selection:

Choose appropriate machine learning algorithms based on the specific requirements of fraud detection, such as decision trees, random forests, gradient boosting, or deep learning models. Consider both supervised and unsupervised learning techniques.

Feature Engineering:

Develop robust feature extraction and selection processes. Identify key features that contribute to distinguishing between legitimate and fraudulent transactions, such as transaction amount, location, time, and user behavior patterns.

Handling Data Imbalance:

Implement techniques to address data imbalance, such as oversampling the minority class (fraudulent transactions), under sampling the majority class (legitimate transactions), or using synthetic data generation methods like SMOTE.

Data Diversity:

Include a wide range of transaction types and behaviors in the dataset to create a comprehensive model that can generalize well across different scenarios.

Data Privacy and Security:

Adhere to data protection regulations such as GDPR and CCPA. Anonymize and encrypt sensitive data to protect user privacy and prevent unauthorized access.

Training Process:

Use a well-structured training process with a clear split between training, validation, and test datasets to avoid overfitting and ensure generalizability.

Cross-Validation:

Apply cross-validation techniques to assess model performance reliably. Use k-fold cross-validation to evaluate how the model performs on different subsets of the data.

Evaluation Metrics:

Employ appropriate metrics for fraud detection, including precision, recall, F1-score, AUC-ROC, and confusion matrix. Focus on metrics that highlight the model's ability to accurately detect fraud while minimizing false positives and false negatives.

Minimal Disruption:

Strive to minimize the impact of fraud detection mechanisms on legitimate transactions to maintain a smooth user experience.

1.5 Objectives

1. Develop models that can precisely identify fraudulent transactions with high accuracy, minimizing both false positives and false negatives.
2. Implement systems capable of analyzing and detecting fraud in real-time to prevent fraudulent transactions from being processed.
3. Minimize the number of legitimate transactions incorrectly flagged as fraudulent to ensure a smooth customer experience.
4. Utilize a wide range of transaction features (e.g., amount, location, time, user behavior) to enhance the model's ability to detect fraud.
5. Create models that can adapt and respond to new and evolving fraud patterns through continuous learning and updates.
6. Ensure the fraud detection system can scale to handle large volumes of transactions efficiently, especially during peak periods.

1.6 Overall Organization of the Thesis

In Chapter 1, a brief explanation has been provided that includes the motivation, problem definition, constraints and objectives of the online payment fraud detection using various machine learning techniques.

In Chapter 2, a through literature survey has been shown highlighting the previous works conducted by various researches along with their shortcomings.

In Chapter 3, the proposed workflow is presented and briefly explained the major design architecture with model discussion.

In Chapter 4, the dataset is discussed along with their detailed features. Moreover, the evaluation metrics and results are also discussed.

In Chapter 5, the conclusion and the future scope is explained in detail.

CHAPTER-2

RECENT LITERATURE

2.1 Recent Literature

The analysis and comprehension of this study involved reviewing several previous studies, leading to the conclusion that numerous issues needed resolution to enhance both efficacy and precision. According to Noor Saleh Alfaiz et al. [1], they proposed the K-Nearest Neighbors under-sampling technique along with the Categorical boosting [Cat Boost] algorithm as the best model for credit card fraud detection. Olawale Adepoju et al. [2] considered exceptionally distorted data on credit card fraud and concluded that the logistic regression algorithm is the best-suited algorithm for credit card fraud detection. Mosa M. M. Megdad et al. [3] concluded in their research paper that the Random Forest classifier is the best classifier for an unbalanced dataset, and the Bagging classifier is the best classifier for a balanced dataset. In a review paper [4], various techniques for detecting fraud in online transactions are listed, offering an understanding of various research papers in the area of fraud detection and prevention. Future research may explore different input and output consideration configurations with machine learning algorithms to identify fraudulent online transactions. The research on "Fraud Detection in Online Payment System using Machine Learning" [5] provides a technique for detecting fraud that combines two types of machine learning algorithms. The authors rely on nuances such as transaction amount, transaction timing, and cardholder information to detect rogue transactions, finding that the SVM method outperforms other techniques. logistic regression is also discussed as a method for solving categorization problems. Another research study [6] explores machine learning algorithms such as SVM, LR, naive bayes, decision tree, and random forest for the same purpose, observing that the Random Forest classifier outperforms other classifiers with the best accuracy.In a separate study [7], the xgboost algorithm, a cluster of decision trees, is highlighted as a method for achieving the best outcomes in fraud detection. Comparisons with other ML algorithms demonstrate that xgboost is faster and more accurate. Similarly, another research study [8] implements the CSO-SVM method, which obtains high performance compared to existing methods including Competitive swarm optimization and Deep convolutional neural network, dilated CNN, and coarse K-neural network, achieving high accuracy. In yet another study [9], various anomaly detection techniques, such as the random forest algorithm and decision tree

classifier, are used to analyze and pre-process datasets for Prompt Corrective Action modified credit card Transaction data. The models are compared and evaluated based on training and testing accuracy, with the decision tree classifier performing better on training accuracy while random forest demonstrates better testing accuracy. Another research study [10] employs machine learning techniques like random forest and logistic regression to improve fraud detection in credit cards. This survey reviews the work by M. Mohana et al. [11] on credit card fraud detection using neural network autoencoders, highlighting key methodologies, findings, and implications for the field. This survey examines the study by G. K. Arun and P. Rajesh [12] which presents a credit card fraud detection model integrating metaheuristic feature selection with deep learning. This survey reviews the work by F. A. Ghaleb et al.[13] on an innovative approach combining GANs, Random Forest, and ensemble synthesized minority oversampling to address credit card fraud detection challenges. This survey reviews the study by P. K et al. [14], which proposes a CNN-BiGRU-A approach for comparative evaluation of fraud detection in online payments. This survey reviews the study by B. Bayram, B. Köroğlu, and M. Gönen [15], which investigates the use of IGBT for improving fraud detection and addressing concept drift in credit card transactions. This survey reviews the work by A. Alshammari et al. [16], which explores the use of Big Data Analytics and various machine learning models for detecting credit card fraud. This survey reviews the study by C. C. Meng et al. [17] on utilizing TabNet for credit card fraud detection.This survey reviews the work by A. A. Taha and S. J. Malebary [18], which explores the use of an optimized Light Gradient Boosting Machine (LightGBM) for detecting credit card fraud. This survey reviews the study by M. A. Alamri and M. A. Ykhlef [19], which presents a machine learning-based framework for detecting credit card anomalies and fraud. This survey reviews the study by P. A. Jadhav et al. [20], which focuses on the identification of fraudulent credit card transactions using various machine learning algorithms.

Table 2.1: Summary of the existing approaches

REFERENCES ENO	DATASET	METHODOLOGY	PERFORMANCE
[1]	Credit Card Transaction Data	K-Nearest Neighbours, CatBoost	Accuracy-98%
[2]	Distorted Credit Card Fraud Data	Logistic Regression	Accuracy-97%
[3]	Unbalanced & Balanced Dataset	Random Forest, Bagging Classifier	Accuracy-95%
[4]	Various Online Transaction Data	Review of Various Techniques	Accuracy-94%
[5]	Online Payment System Data	SVM, Logistic Regression	Accuracy-96%
[6]	Various Machine Learning Datasets	SVM, LR, Naive Bayes, Decision Tree, Random Forest	Accuarcy-95%
[7]	Credit Card Fraud Data	XGBoost	Accuracy-98%
[8]	Competitive Swarm Optimization Data	CSO-SVM	Accuracy-90%
[9]	Prompt Corrective Action Modified Data	Random Forest, Decision Tree	Accuracy-96%
[10]	Credit Card Fraud Data	Random Forest, Logistic Regression	Accuarcy-92%
[11]	Neural Network Autoencoders Data	Neural Network Autoencoders	Accuracy-97%
[12]	Metaheuristic Feature Selection Data	Metaheuristic Feature Selection, Deep Learning	Accuracy-98%
[13]	GANs and Minority	GANs, Random Forest, Ensemble Synthesized	Accuracy-95%

	Oversampling Data	Minority Oversampling	
[14]	CNN-BiGRU-A Data	CNN-BiGRU-A	Accuracy-96%
[15]	Incremental Gradient Boosting Trees Data	Incremental Gradient Boosting Trees	Accuracy-92%
[16]	Big Data Analytics Data	Big Data Analytics, Various ML Models	Accuracy-98%
[17]	TabNet Data	TabNet	Accuracy-93%
[18]	Light Gradient Boosting Machine Data	Optimized LightGBM	Accuracy-92%
[19]	Machine Learning-Based Anomaly Detection Data	Machine Learning-Based Framework	Accuracy-95%
[20]	Various Machine Learning Algorithms Data	Various ML Algorithms	Accuracy-94%

2.2. Limitations:

1. High-quality, labeled datasets of fraudulent and non-fraudulent transactions are essential for training ML models. However, obtaining such data can be challenging due to privacy concerns and the relatively low incidence of fraud compared to legitimate transactions.
2. Fraudulent transactions typically represent a small fraction of total transactions, leading to imbalanced datasets. This imbalance can bias models towards predicting non-fraudulent behavior, reducing their ability to detect actual fraud.
3. Fraudsters constantly change their techniques to evade detection. ML models trained on historical data may struggle to adapt to new types of fraud, necessitating frequent retraining and updates to maintain effectiveness.
4. Novel fraud schemes that haven't been seen before pose a significant challenge, as the model lacks historical examples to learn from.
5. Identifying the right features that accurately represent fraud-related patterns is complex and often requires domain expertise. Poor feature selection can significantly reduce model performance.
6. ML models can overfit to the training data, especially if too many irrelevant features are included. This makes them less effective at generalizing to new, unseen data.

CHAPTER-3

PROPOSED SYSTEM

3.1 Proposed Workflow:

The architecture of the proposed work is shown in Figure 3.1, that shows the identification of fraud in online payments. The dataset, which has been taken from Kaggle, includes transactional data. Key characteristics are extracted from the gathered dataset and preprocessed thereafter. After that the classification methods(SVM, LR, DT and RF) are applied. Following the implementation of different methods, the accuracy of the fraud detection is determined.

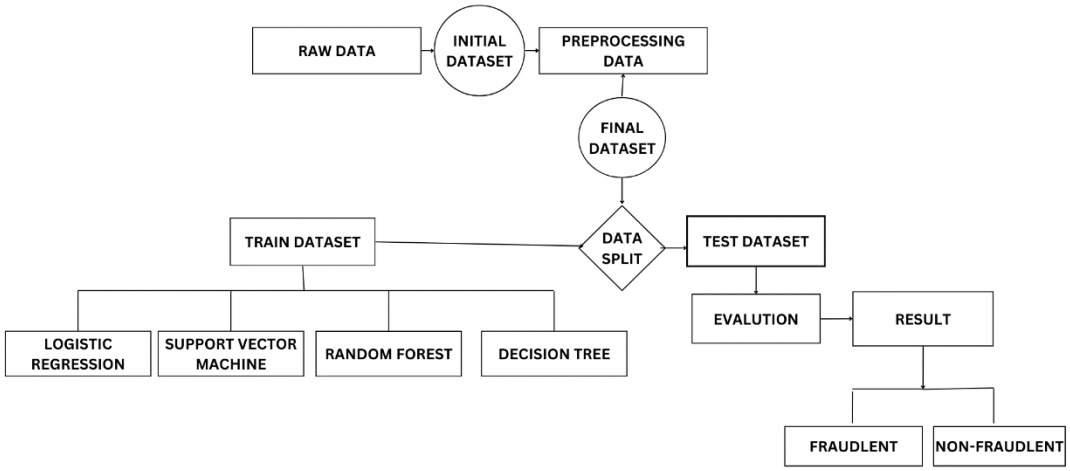


Figure 3.1 System Workflow.

Subsequently, the extracted features undergo a preprocessing phase, which involves cleaning the data, handling missing values, and normalizing or standardizing the data to ensure consistency and accuracy in the subsequent analysis. This preprocessing step is crucial as it prepares the data for effective model training. The implementation of these different classification methods is carried out systematically, allowing for a comparative analysis of their performance. After applying each method, the accuracy of the fraud detection is determined and evaluated. Accuracy metrics, such as precision, recall, and F1-score, are calculated to assess the effectiveness of each classifier in identifying fraudulent transactions

3.2 Discussion of Proposed Workflow Diagram:

In the Figure 3.2 that shows to develop an effective online payment fraud detection system using a Decision Tree, the initial step involves gathering a substantial dataset comprising records of online transactions. This dataset should encompass a wide array of features pertinent to identifying fraudulent activities, such as transaction amounts, timestamps, locations, merchant details, and user behavior patterns, along with labels indicating whether each transaction is legitimate or fraudulent. Once collected, the data must be divided into distinct testing and training sets. This division is crucial for assessing the machine learning model's performance and ensuring it generalizes well to unseen data.

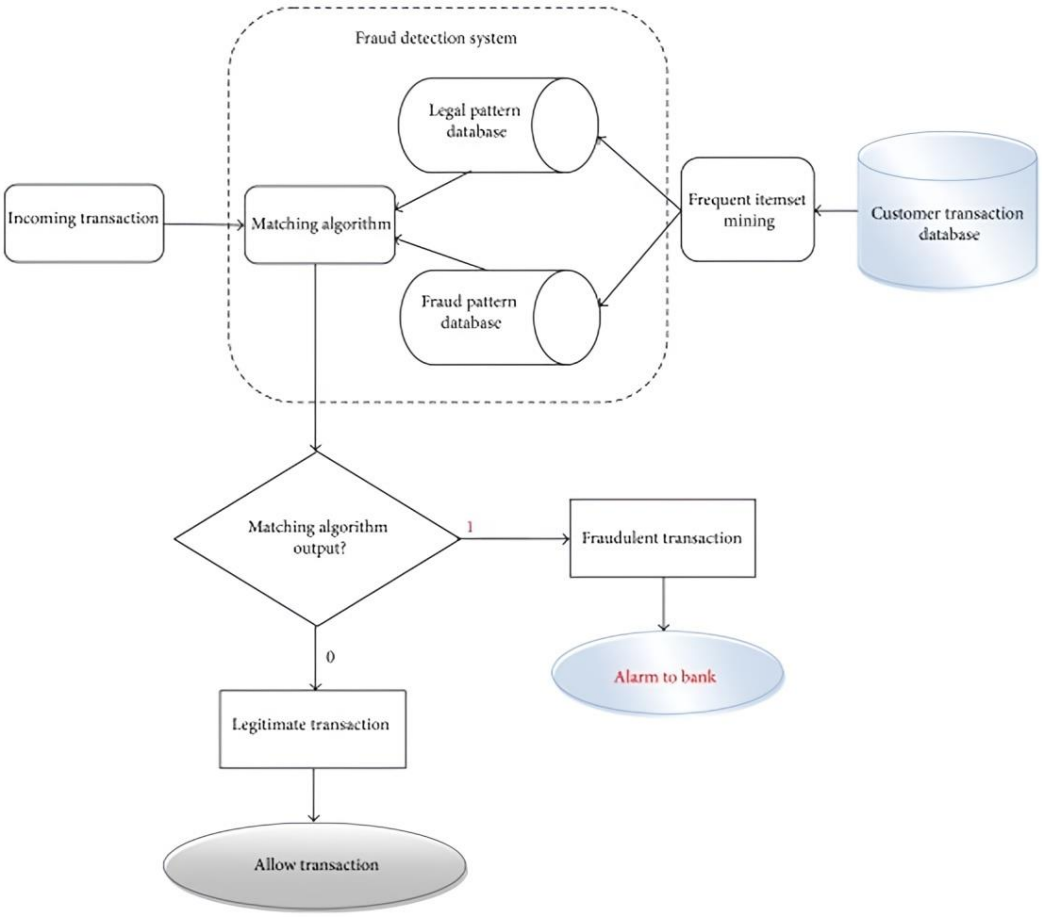


Figure 3.2 Architecture

Before applying any machine learning algorithms, it is essential to preprocess the raw data. This involves employing techniques such as category encoding and scaling. Category encoding transforms categorical variables into a numerical format suitable for the algorithm, while scaling ensures that numerical features are on a comparable scale,

preventing any single feature from disproportionately influencing the model. With the data preprocessed, the next step is to train the model using appropriate methods. For a Decision Tree, this means fitting the model to the training data by learning the decision rules that best separate fraudulent transactions from legitimate ones.

3.3. Data Analysis:

Data analysis is essential for identifying fraudulent online payment transactions. With the use of machine learning techniques, banks and other financial institutions can modify the necessary defenses against these scams. Large sums of money are being invested by numerous companies and organizations in the creation of these machine learning systems to ascertain whether a particular transaction is fraudulent. These machine learning techniques to help them detect frauds and protect their clients who can be vulnerable to them and occasionally suffer losses as a result.

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
0	1	PAYMENT	9839.64	C1231006815	170136.0	160296.36	M1979787155	0.00	0.00	0	0
1	1	PAYMENT	1864.28	C1666544295	21249.0	19384.72	M2044282225	0.00	0.00	0	0
2	1	TRANSFER	181.00	C1305486145	181.0	0.00	C553264065	0.00	0.00	1	0
3	1	CASH_OUT	181.00	C840083671	181.0	0.00	C38997010	21182.00	0.00	1	0
4	1	PAYMENT	11668.14	C2048537720	41554.0	29885.86	M1230701703	0.00	0.00	0	0
...	...	...	...	...	...	...	...	...	...	...	...
195	1	CASH_OUT	210370.09	C2121995675	0.0	0.00	C1170794006	1442298.03	22190.99	0	0
196	1	CASH_OUT	36437.06	C2120063568	0.0	0.00	C1740000325	154606.00	1363368.51	0	0
197	1	CASH_OUT	82691.56	C1620409359	0.0	0.00	C248609774	657983.89	6453430.91	0	0
198	1	CASH_OUT	338767.10	C691691381	0.0	0.00	C453211571	544481.28	3461666.05	0	0
199	1	CASH_OUT	187728.59	C264978436	0.0	0.00	C1360767589	394124.51	2107965.39	0	0

Figure 3.3: Dataset

In the Figure 3.3 that shows the online platform “kaggle” provided the data set for the study. Real-time data sets are difficult to gather due to privacy concerns; however, a large enough data collection was made to carry out the research. There are 11 columns and 1048576 records in the data set. Type (payment type), amount,”nameOrig” (customer initiating the transaction), “oldbalanceOrig" (balance before the transaction), ”newbalanceOrig” (balance after the transaction), ”nameDest” (recipient transaction), 5 ”pldbalanceDest” (initial recipient balance prior to the transaction),”newbalanceDest” (the new balance recipient after the transaction), and isFraud (0 if the transaction is legitimate and 1 if it is fraudulent) are among the attributes included in this data set.

3.3.1. Analyzing Missing Values:

Preprocessing the data that was downloaded from the dataset is crucial before utilizing utilizing it in the model. The next action is to examine our dataset to see if any values missing. Figure 3.4 illustrates that our dataset has no missing values.

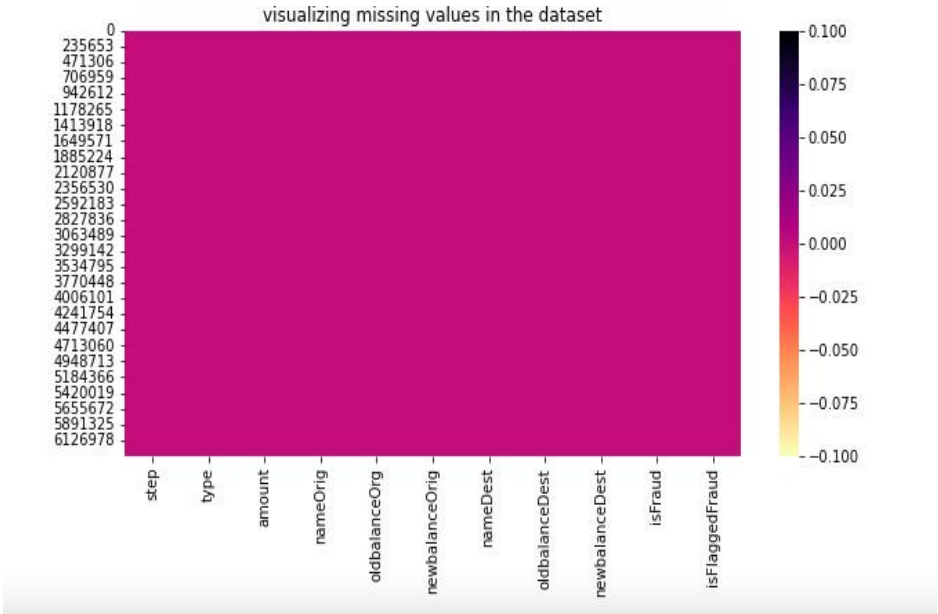
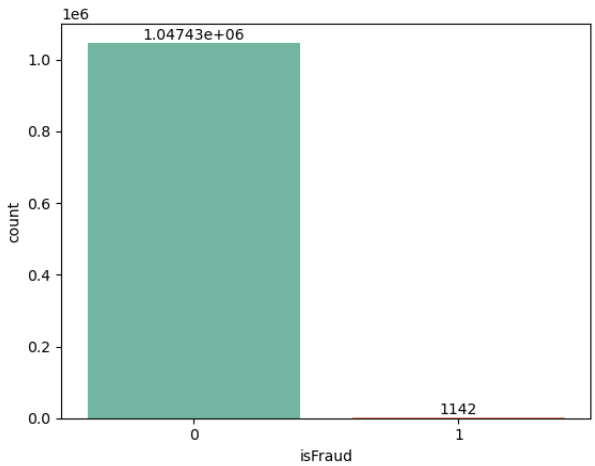


Figure 3.4: Missing Values

3.3.2 Target Distribution Investigation:

It is evident from examining the distribution of our objective characteristic, "isfraud," that our data exhibits significant class imbalance.

Figure 3.5 illustrates that, out of 6354407 transactions, only 8213 are reported as fraudulent. We would address this disparity across classes in subsequent phases to ensure that it has no adverse effects on our machine learning models.



Figures 3.5: Imbalance data

3.3.3. Relation between the characteristics:

In the Figure 3.6 that shows in machine learning, correlation refers to the statistical relationship between two variables. It indicates the degree to which changes in one variable are associated with changes in another. Correlation analysis is commonly used to understand the relationship between features (independent variables) and the target variable (dependent variable) in a dataset.

Despite the abundance of features in our data, not all of them are contributing to the feature we are aiming for. Reduced the number of significant features that can assist us in predicting our target feature with the use of the correlation matrix. But in order to compare the models' output, we will retrain the models using all of the attributes that were previously rejected.

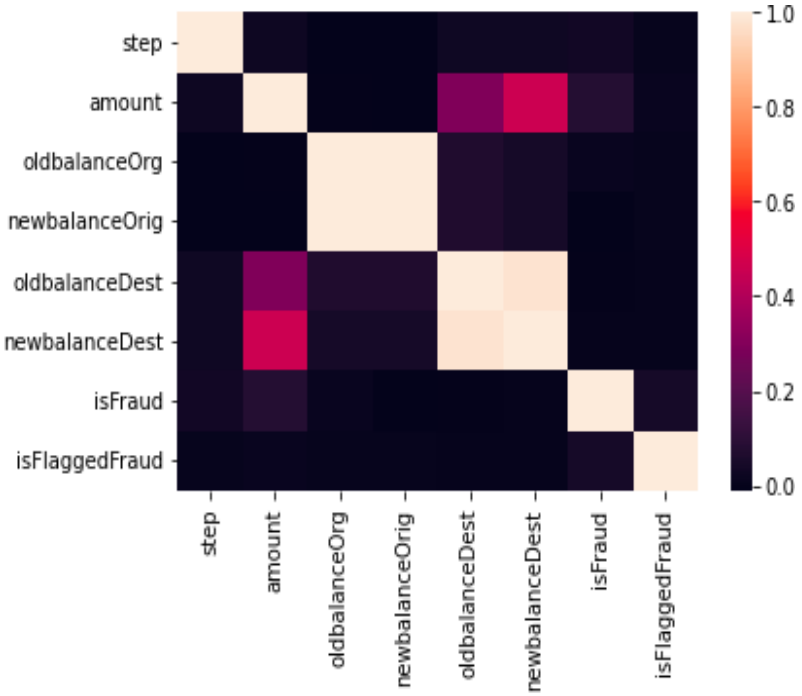


Figure 3.6: Correlation

3.3.4 Data preparation:

For the machine learning model to give accurate, high-quality results, the data used totrain and test it should be well-prepared. One of the most important steps in data mining is getting the data ready. There are many things to consider, such as how to deal with missing data, duplicate values, removing redundant features from data using correlation matrix and feature selection methods, how to deal with the fact that data isn't balanced, etc. The quality of the techniques used to prepare the data has a lot to do with how well machine learning works. If the data is not prepared well, it could take a long time to runthe models and cost a lot of money. Because of all of these things, the most difficult and time-consuming part of the data mining process is getting the data ready.

3.3.5 Feature Selection:

Feature selection is one of the approaches that helps models perform even better after data cleansing and feature correlation analysis. This method is used to eliminate unnecessary variables, which leads to a smaller feature space and could improve the performance ofthe model.

In our dataset two features "namedest" and "nameOrig" are of less significance as compared to other features, however to compare the same we will be running the models without these features and then including these two features.

3.3.6 Handling Class Imbalance

One of the key issues in the field of fraud detection is the class imbalance, which is covered in the sections above. Algorithms for machine learning are created to work optimally when taught on sufficient examples from both classes. In the Figure 3.7 that shows the performance of machine learning models is vulnerable to skewed outcomes and overfitting due to the rarity of fraud transactions within the overall data. For the class samples with lower representation, this could lead to incorrect classification. There are a number of sampling approaches thatcan help with this problem, each with

its own set of benefits and drawbacks. These strategies either focus on oversampling the minority class, under sampling the dominant class, or a mix of the two.

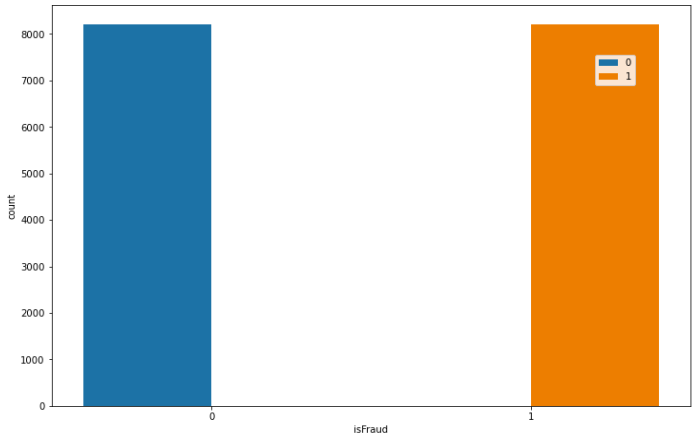


Figure 3.7: Target Feature count after the under sampling

3.4. Modelling Approach:

Modelling is a very important aspect in machine learning. After the final data preparation, which includes steps like handling the class imbalance and feature selection, the proposed models are implemented on the processed or prepared data. The detailed explanation and working of the proposed models are discussed in this section:

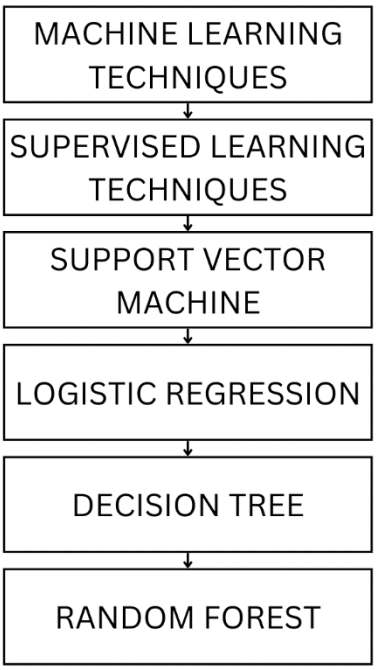


Figure 3.8: Machine learning techniques

3.4.1 Supervised Learning

Supervised learning is a prominent paradigm within machine learning where algorithms are trained on labeled data, enabling them to learn the mapping between input features and output labels. In supervised learning, the algorithm is provided with a dataset containing input-output pairs, where each input is associated with a corresponding output label. The primary objective is to learn a model that can accurately predict the output label for new, unseen inputs based on the patterns observed in the training data.

There are two main types of supervised learning: classification and regression. In classification tasks, the output labels are discrete and categorical, such as class labels (e.g., spam or not spam, dog or cat). The goal is to classify new inputs into one of the predefined classes. In regression tasks, the output labels are continuous numerical values, such as price or temperature. The goal is to predict a numerical value based on the input features. Supervised learning algorithms employ various techniques, such as decision trees, support vector machines, neural networks, and ensemble methods, to learn complex relationships between inputs and outputs.

Supervised learning finds applications in a wide range of domains, including image and speech recognition, medical diagnosis, financial forecasting, and recommendation systems. By leveraging labeled data to train predictive models, supervised learning enables machines to make informed decisions and predictions in real-world scenarios, driving advancements in technology and improving decision-making processes across various industries.

3.4.1. Support Vector Machine:

SVM stand as a pivotal class of supervised learning algorithms renowned for their proficiency in both classification and regression tasks. These models excel particularly in high-dimensional spaces, demonstrating a remarkable aptitude for handling both linear and non-linear data distributions. At the core of SVM lies the notion of discovering the optimal hyperplane that best segregates data points belonging to different classes.

This hyperplane, while a line in two-dimensional spaces, extends to planes or hyperplanes in higher dimensions. The SVM algorithm endeavors to maximize the

margin, defined as the distance between the hyperplane and the nearest data points from either class, termed support vectors. Such support vectors significantly influence the position and orientation of the hyperplane. When faced with non-linearly separable data, SVMs utilize kernel functions to transform the data into higher-dimensional spaces where linear separation becomes viable.

Hyperplane:

A hyperplane is a flat affine subspace of one dimension less than its ambient space. In the context of SVM, it is used to separate data points into different classes. Mathematically, a hyperplane in an n-dimensional space is defined by the Equation 3.1:

$$w.\ x + b = 0 \tag{3.1}$$

Decision Function:

By transferring the points from a simple 2D dimension to a higher dimension, the machine learning SVM algorithm finds the plane that distinguishes the highest number of points. For this, we have employed the regularization parameter along with the kernel, which is the sigmoid function Equation 3.2:

$$f(x) = \text{sign}\ (w.\ x + b) \tag{3.2}$$

Prediction:

The SVM prediction function evaluates the new transaction x based on its similarity to the support vectors from the training data. The weighted sum of these similarities, adjusted by the class labels and Lagrange multipliers, determines the position of x relative to the decision boundary. The sign function Equation 3.3 then classifies x as either non-fraudulent (+1) or fraudulent (-1), providing a robust method for detecting fraud in online transactions.

$$f(\mathbf{x}) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b\right) \tag{3.3}$$

Fraud detection often involves analyzing complex patterns and relationships between various features. SVM's ability to handle high-dimensional data makes it well-suited for this task.

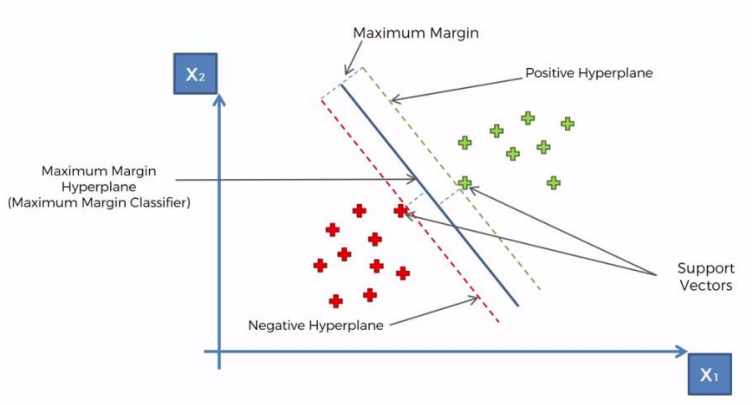


Figure 3.9: Support Vector Machines

3.4.2. Logistic Regression:

Logistic Regression is a fundamental machine learning algorithm used primarily for binary classification tasks, where the goal is to predict the probability that an instance belongs to a particular class. Despite its name, logistic regression is a classification algorithm rather than a regression algorithm. At its core, logistic regression models the relationship between a set of independent variables and a binary dependent variable by estimating the probability of the dependent variable taking on a specific value.

Unlike linear regression, logistic regression utilizes the logistic function (also known as the sigmoid function) to model the probability that a given input belongs to the positive class. This sigmoid function maps any real-valued input to the range [0, 1], effectively transforming the output into a probability score. The model's parameters, including the weights assigned to each feature and the bias term, are learned through optimization techniques such as gradient descent, with the objective of maximizing the likelihood of the observed data. Logistic Regression is used to model the probability of a binary outcome based on one or more predictor variables (features). The logistic function (or sigmoid function) maps the predicted values to probabilities between 0 and 1. In online payment fraud detection, Logistic Regression can be particularly effective due to its simplicity, interpretability, and efficiency. The model can be trained on features such as transaction amount, transaction time, user behavior metrics, and more to predict the probability of a transaction being fraudulent.

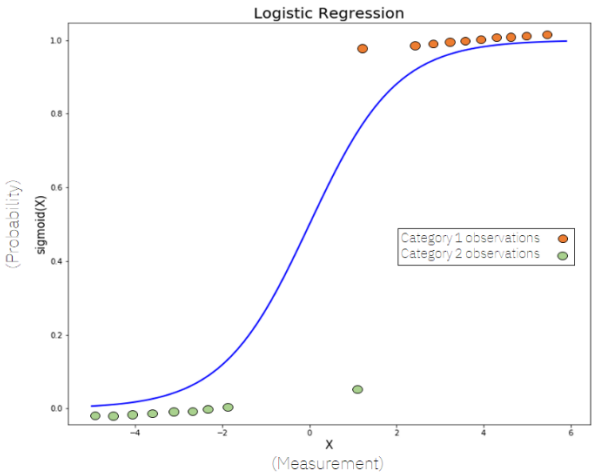


Figure 3.10: Logistic Regression

The method that uses previously trained data to classify data as either fraudulent or not is called logistic regression. Regression is the name given to the process that separates one class from another by drawing a line. The predictions of our model are '0' for no fraud and '1' for fraud. In light of this, the data is divided into two classes: class 2, which is not fraudulent, and class 1, which is fraudulent.

3.4.3 Random Forest

Random Forest is a versatile and powerful ensemble learning algorithm widely used in machine learning for both classification and regression tasks. It operates by constructing a multitude of decision trees during training and outputs the class that is the mode of the classes (classification) or the mean prediction (regression) of the individual trees. Each decision tree in the random forest is trained on a bootstrap sample of the training data, and at each node of the tree, a random subset of features. This randomness injected into the training process helps mitigate overfitting and improves the generalization performance of the model. During prediction, each tree in the forest independently outputs its prediction, and the final prediction is determined through a majority vote (classification) or averaging (regression) of the individual tree predictions. Random Forest is an ensemble learning method primarily used for classification and regression tasks. It operates by constructing multiple decision trees during training and outputting the mode of the classes (classification) or mean prediction (regression) of the individual trees.

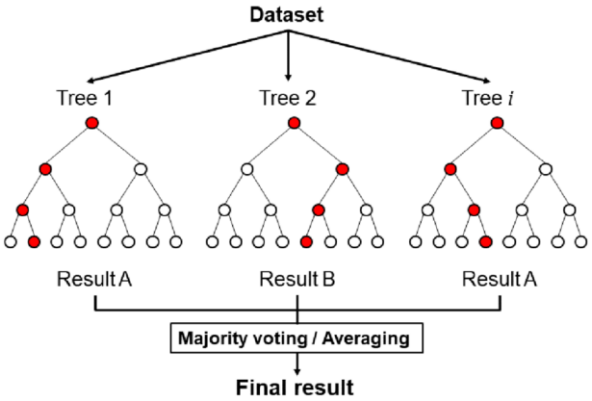


Figure 3.11: Random Forest

Generally aware that a forest is made up of trees. A collection of decision trees is what random forests are, too. The complete dataset is divided into subsets by this algorithm, which then trains on each subset a predetermined number of times. 500 trees have been used, and we have trained on them 100 times. We were therefore able to achieve the maximum accuracy of 0.999453%.

3.4.4. Decision Tree:

Decision Trees are a fundamental machine learning algorithm used for both classification and regression tasks. They operate by recursively partitioning the input space into regions, guided by a series of decision rules inferred from the training data. At each node of the tree, the algorithm selects the feature that best splits the data into homogenous subsets according to some criterion, such as Gini impurity or information gain. This process continues until a stopping criterion is met, such as reaching a maximum tree depth or no further improvement in purity.

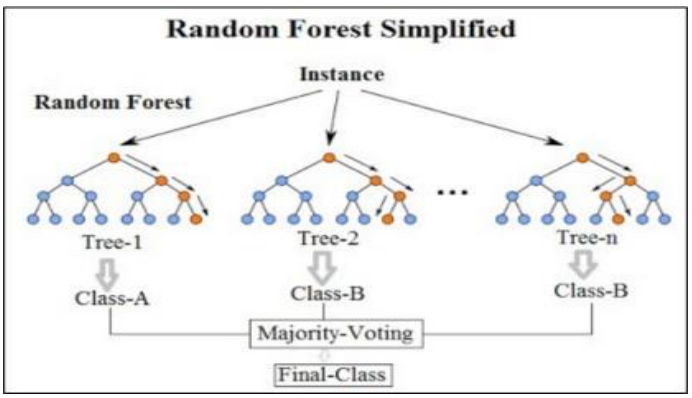


Figure 3.12: Decision Tree

Entropy:

Entropy is a measure of impurity or uncertainty in a dataset. In the context of a binary classification problem (such as classifying transactions as fraudulent or non-fraudulent), entropy quantifies the level of disorder or randomness in the dataset. It is defined mathematically as Equation 3.5:

$$H(S) = -p_1 \log_2(p_1) - p_2 \log_2(p_2) \tag{3.5}$$

Information Gain:

Information Gain (IG) measures the reduction in entropy after a dataset is split on a particular feature. It helps in determining which feature provides the most information about the classification in Equation 3.6:

$$IG(S, A) = H(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} H(S_v) \tag{3.6}$$

Gini Impurity:

Gini Impurity is another measure of impurity used in decision trees, similar to entropy but calculated differently. In the Equation 3.7It measures the probability of incorrectly classifying a randomly chosen element if it was labeled according to the distribution of labels in the dataset.

$$G(S) = 1 - \sum_{i=1}^{c} p_i^2 \tag{3.7}$$

Gini Gain:

The formula for Gini Gain is Equation 3.8:

$$GiniGain(S, A) = G(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} G(S_v) \tag{3.8}$$

One of the key advantages of Decision Trees is their interpretability. The resulting tree structure provides clear insights into the decision-making process, allowing users to understand and interpret the model's predictions easily. Additionally, Decision Trees can handle both numerical and categorical data, making them versatile for various types of datasets. They are also robust to outliers and missing values, as the splitting process is based on purity measures rather than absolute values.

CHAPTER-4

EXPERIMENTAION AND RESULT ANALYSIS

4.1 Dataset Details

For the purpose of detecting fraud, this work references the dataset from the Kaggle repository titled "Online Payments Fraud Detection." The considered dataset is fully unbalanced, implying there are far fewer fraudulent transactions than non-fraudulent ones. To address this issue, various machine learning techniques (such as support vector machines, logistic regression, random forest, and decision tree algorithms) have been employed in this work, yielding superior results even with unbalanced datasets.

4.2 Performance Evaluation metrics

The main goal of this research is to use supervised machine learning techniques and see if our proposed method improves the model's performance more than other state-of-the-art methods. we chose metrics like recall, specificity, F1-score, AUC score, AUC-ROC curve, and the geometric mean of recall and specificity so that we could compare how well our models worked.

And also compares the results based on the confusion matrixas in this case better to compare the True Positive and True negatives and decide based on that that how much accuracy is achieved in our models. In the Figure 4.1 that shows the confusion matrixis explained below:

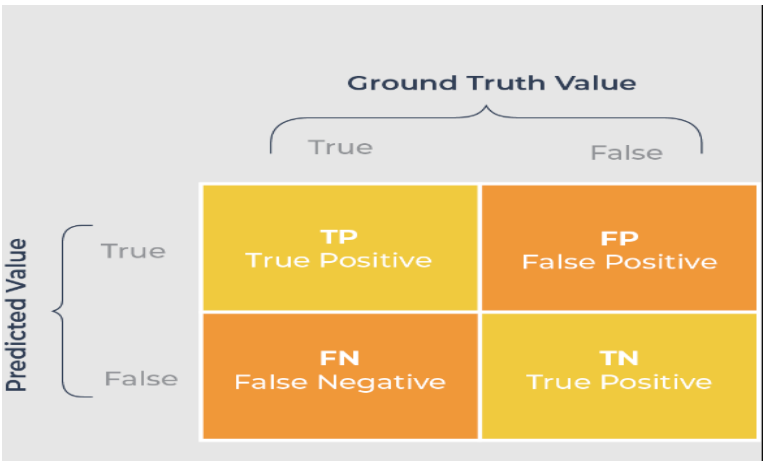


Figure 4.1: Confusion Matrix

True Positive (TP): It shows that the given model has done a good job of figuring out non-fraudulent cases as non-fraud (positive).

False Positive (FP): It shows that the model didn't get the prediction right, fraudulent cases were identified as non-fraud (positive).

False Negative (FN): It shows that the model didn't get the prediction right, non-fraudulent cases were identified as fraudulent (negative).

True Negative (TN): It shows that the model has been able to accurately predict fraudulent cases as fraudulent (negative).

Recall:

Recall measures the proportion of correctly classified tumor pixels among all actual tumor pixels. In the Equation 4.1 it quantifies the number of true positive predictions out of all actual positive pixels in the ground truth.

$$\text{Recall/Sensitivity} = \frac{TP}{TP + FN} \tag{4.1}$$

Precision:

Precision is a metric used in binary classification to measure the accuracy of the positive predictions. In the Equation 4.2 it answers the question: "Of all the instances that were predicted as positive, how many were actually positive?" Precision is particularly useful when the cost of false positives is high.

$$\text{Precision/Specificity} = \frac{TP}{TP + FP} \tag{4.2}$$

F1-Score:

The F1 score is the harmonic mean of precision and recall, providing a balance between the two metrics. In the Equation 4.3 it represents the model's accuracy in terms of both precision and recall.

$$\text{F1-Score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \tag{4.3}$$

4.3 Discussions:

Online payment fraud detection utilizing machine learning techniques is a multifaceted endeavor that involves careful consideration of hyperparameters, insightful visualization through plots, and rigorous evaluation of model performance.

4.3.1 Hyper Parameters:

Here's a table summarizing the hyperparameters for Support Vector Machine (SVM), Logistic Regression (LR), Decision Trees (DT), and Random Forests (RF):

These hyperparameters can be tuned using techniques like grid search or randomized search to find the optimal combination for the given dataset and problem.

Table 4.1: Hyper Parameters

Algorithm	Hyperparameter	Values
Support Vector Machine (SVM)	C	0.1, 1, 10, 100
	kernel	rbf, linear
	degree	3
Logistic Regression (LR)	C	0.01, 0.1, 1, 10, 100
	penalty	12
	Solver	lbfgs, liblinear, saga
Decision Trees (DT)	criterion	gini, entropy
	max_depth	10,20,30
	min_samples_split	2,10,20
	min_samples_leaf	1,2,4
Random Forests (RF)	n_estimators	100,200,500
	max_depth	10,20,30
	min_samples_split	2,10,20
	min_samples_leaf	1,2,4
	max_features	sqrt,log2

4.3.2 Plots and Confusion Matrix:

To create meaningful plots for online payment fraud detection using machine learning, we can visualize various aspects of the dataset, model performance, and results. Below are examples of plots we can generate:

Data Distribution Plots:

Transaction Amount Distribution: In the Figure 4.1 plotting the distribution of transaction amounts can help identify any unusual patterns or outliers. Figure 4.3 Fraudsters may attempt to make small transactions to test stolen credit card information before making larger purchases.

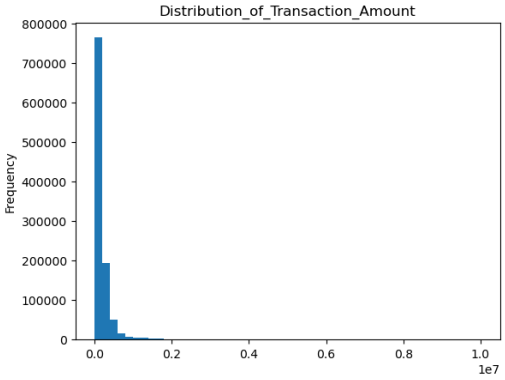


Figure 4.2: Transaction Amount

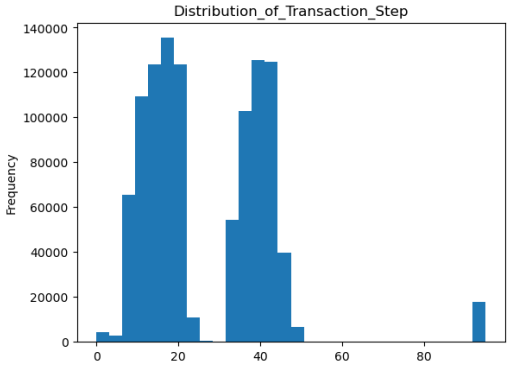


Figure 4.3: Transaction Step

Transaction Frequency Distribution: In the Figure 4.4 analyzing the distribution of transaction frequencies can reveal any abnormal spikes or patterns in transaction activity. In the Figure 4.5 For instance, a sudden increase in transactions from a particular account could indicate fraudulent activity.

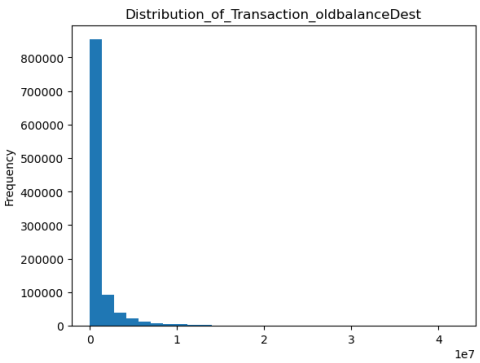


Figure 4.4: Old Balance

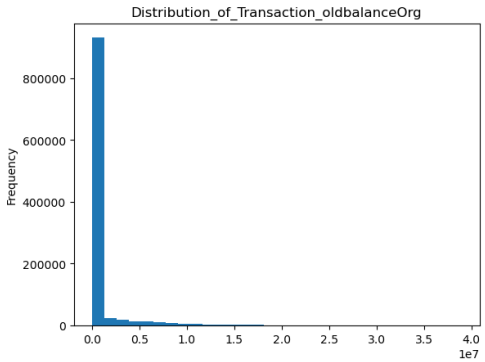


Figure 4.5: New Balance

Confusion Matrix:

In the context of online payment fraud detection using machine learning, a confusion matrix is a crucial tool for evaluating the performance of a fraud detection model. It provides a detailed breakdown of the model's predictions compared to the ground truth labels, distinguishing between different types of classification outcomes.

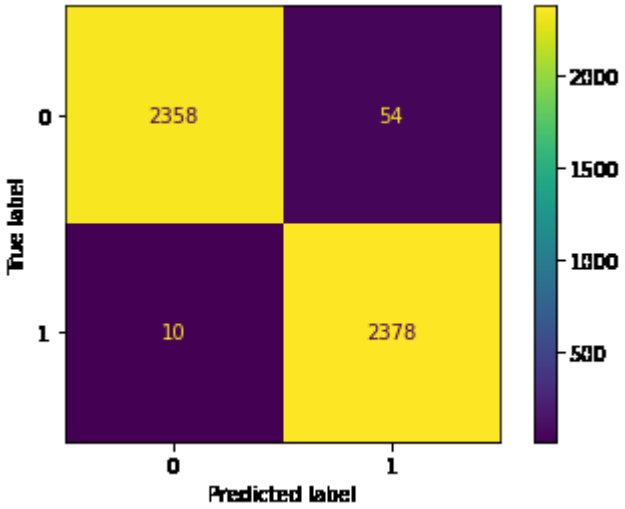


Figure 4.6: Confusion Matrix on Proposed Model

4.3.3 Comparative Analysis:

Assessment metrics are employed to gauge how effectively an AI or factual model is doing. Any AI models or calculations used in a project must be evaluated. There are many evaluation tools available to test a model.

We looked at accuracy and calculated values for each algorithm to determine which one was best suited for detecting fraud. Then, we compared each algorithm's values to those of the other algorithms to determine which one was the best fit.

The frequency with which the model predicts the right value is known as accuracy, and it is determined by dividing the total number of correct predictions by the total number of predictions generated by the classifier. Enhanced data integration from various sources, such as social media and customer support interactions, will offer a more comprehensive view of transaction risk, improving the context-awareness of fraud detection systems. Continuous learning mechanisms will ensure that systems remain effective against evolving threats by regularly updating models with new data and feedback.

Table 4.2: Comparative Analysis

Classifier	Accuracy	Precision	Recall	F1Score
Support vector machine	55.68%	86.18%	89%	90%
Logistic Regression	95.45%	96.3%	97%	98%
Random Forest	99.12%	80%	97%	78%
Decision Tree	99.96%	98%	98%	99%

After analyzing the outcomes of every supervised machine learning method previously described, it has been determined that the Decision Tree algorithm, which has an accuracy of 99.96%, is the most appropriate for detecting online transaction fraud. The performance of the Decision Tree model is evaluated using accuracy, as indicated in TABLE 4.I. It is observed that the Random Forest and Decision Tree algorithms produce equal percentages. The evaluation results demonstrate the robustness and reliability of the Decision Tree algorithm in detecting fraudulent payments. Nevertheless, after examining additional performance metrics, it is evident that the Decision Tree produces better outcomes

4.3.4 Sample Output:

In the realm of online payment fraud detection using machine learning, the confusion matrix serves as a fundamental tool for evaluating model performance. For instance, consider a model that processes a batch of transactions and predicts whether each is fraudulent or legitimate. The confusion matrix would categorize the outcomes into four groups: true positives (fraudulent transactions correctly identified), false positives (legitimate transactions incorrectly flagged as fraudulent), true negatives (legitimate transactions correctly identified), and false negatives (fraudulent transactions missed by the model).

These metrics reveal the model's performance in real-world conditions, highlighting its strengths and areas for improvement. By interpreting the confusion matrix and associated metrics, organizations can refine their fraud detection models to better balance false positives and false negatives, ensuring robust and reliable protection against fraudulent activities in online payment systems.

```
model.predict([[3,181,181,0]])
```

```
C:\Users\gangi\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarning:

X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
```

```
array(['Fraud'], dtype=object)
```

```
model.predict([[2,400.00,500.00,100.00]])
```

```
C:\Users\gangi\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarning:

X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
```

```
array(['No_Fraud'], dtype=object)
```

```
model.predict([[2,3050.55,11929.01,56886.00]])
```

```
C:\Users\gangi\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarning:

X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
```

```
array(['No_Fraud'], dtype=object)
```

Figure 4.7: Prediction Output

CHAPTER-5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

In this work, various machine learning algorithms, like RF, SVM, LR and DT, were implemented to predict whether a particular transaction is fraudulent or not. To improve the performance of the models, various techniques such as handling class imbalance, feature selection was used in order to extract the most relevant data. A comparison among all the adopted approaches has been presented thoroughly to demonstrate their effectiveness in solving the considered issue. In this work, the Random Forest produces a lower accuracy of 99.12%, while the Decision Tree provides the best accuracy of 99.96%. From the comparison it has been observed that the decision tree algorithm method handles the issue more effectively than other algorithms.

5.2 Future Scope

Future work for this paper will involve exploring additional machine learning algorithms for detection and building a user-friendly interface for inputting new data and detecting whether a transaction is fraudulent. Additionally, a deep learning approach can be applied to the same dataset to obtain results more accurately and quickly. The future of online payment fraud detection is poised to evolve significantly, driven by advancements in technology, increasing sophistication of fraud tactics, and the growing adoption of digital payments. Advanced machine learning and artificial intelligence, particularly deep learning and neural networks, will enhance the accuracy and efficiency of fraud detection systems by better understanding complex patterns in transaction data. Real-time fraud detection will become more crucial as the volume and speed of online transactions increase, with technologies like in-memory computing enabling instantaneous analysis and flagging of suspicious transactions.

REFERENCES

[1] Noor Saleh Alfaiz and Suliman Mohamed Fati "Enhanced Credit Card Fraud Detection Model Using Machine Learning ", Electronics 2022,11(4), 662; mdpi.com/2079-9292/11/4/662.

[2] Olawale Adepoju, Julius Wosowei, Shiwani lawte, Hemaint Jaiman "Comparative Evaluation Of Credit Card Fraud Detection Using Machine Learning Techniques", 2019 Global Conference for Advancement in Technology (GCAT)

[3] Mosa M. M. Megdad, Bassem S. Abu-Nasser and Samy S. Abu-Naser "Fraudulent Financial Transactions Detection Using Machine", 2022 International Journal of Academic Information Systems Research (IJAISR).

[4] Online Transaction Fraud Detection System Based on Machine Learning by Virjanand, RajkishanBharti, Shubham Chauhan, and Suraj Pratap Singh-2022

[5] Lochan, S., et al. "Online Payment Fraud Detection Using Machine Learning." 2023 International Conference on Computational Intelligence for Information, Security and Communication Applications (CIISCA). IEEE, 2023.

[6] D. Aladakatti, G. P, A. Kodipalli and S. Kamal, "Fraud detection in Online Payment Transaction using Machine Learning Algorithms," 2022 International Conference on Smart and Sustainable Technologies in Energy and Power Sectors (SSTEPS), Mahendragarh, India, 2022, pp. 223-228, doi: 10.1109/SSTEPS57475.2022.00063.

[7] U. Siddaiah, P. Anjaneyulu, Y. Haritha and M. Ramesh, "Fraud Detection in Online Payments using Machine Learning Techniques," 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 2023, pp. 268-273, doi: 10.1109/ICI-CCS56967.2023.10142404.

[8] A. Gajakosh, R. A. Reddy, M. Mundher adnan, G. Rajalaxmi and P. R, "Fraud Detection in Credit Card Using Competitive Swarm Optimization

with Support Vector Machine," 2024 International Conference on Distributed Computing and Optimization Techniques (ICDCOT), Bengaluru,India, 2024, pp. 1-4, doi: 10.1109/ICDCOT61034.2024.10515953.

[9] A. Gupta, K. Singh, N. Sharma and M. Rakhra, "Machine Learning For Detecting Credit Card Fraud," 2022 IEEE North Karnataka Subsection Flagship International Conference (NKCon), Vijaypur, India, 2022, pp. doi: 10.1109/NKCon56289.2022.10126861.

[10] S. V. J. B. Gracia, J. G. Ponsam, S. Preetha and J. Subhiksha, "Payment fraud detection using machine learning techniques," 2021 4th International Conference on Computing and Communications Technologies (ICCCT), Chennai, India, 2021, pp. 623-626, doi: 10.1109/IC-

[11] CCT53315.2021.9711887.M. Mohana, K. Kumaran, N. Nandhini, N. Ananthi, J. Naresh and K. R. Madhavan, "Credit Card Fraud Detection Using Neural Network Auto Encoders," 2022 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI), Chennai, India, 2022, pp. 1-7, doi: 10.1109/ACCAI53970.2022.9752560.

[12] G. K. Arun and P. Rajesh, "Design of Metaheuristic Feature Selection with Deep Learning Based Credit Card Fraud Detection Model," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), Coimbatore, India, 2022, pp. 191-197, doi: 10.1109/ICAIS53314.2022.9742937.

[13] F. A. Ghaleb, F. Saeed, M. Al-Sarem, S. N. Qasem and T. Al-Hadhrami, "Ensemble Synthesized Minority Oversampling-Based Generative Adversarial Networks and Random Forest Algorithm for Credit Card Fraud Detection," in *IEEE Access*, vol. 11, pp. 89694-89710, 2023, doi: 10.1109/ACCESS.2023.3306621.

[14] P. K, M. S. Giridhar, A. Tatia, T. M. Joshi, S. Pal and U. S. Aswal, "Comparative Evaluation of Fraud Detection in Online Payments Using CNN-BiGRU-A Approach," 2023 International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), Erode, India, 2023, pp. 105-110, doi: 10.1109/ICSSAS57918.2023.10331745.

[15] B. Bayram, B. Köroğlu and M. Gönen, "Improving Fraud Detection

and Concept Drift Adaptation in Credit Card Transactions Using Incremental Gradient Boosting Trees," 2020 19th IEEE International Conference on Machine Learning and Applications (ICMLA), Miami, FL, USA, 2020, pp. 545-550, doi: 10.1109/ICMLA51294.2020.00091

[16] A. Alshammari, R. Alshammari, M. Altalak, K. Alshammari and A. Alhakamy, "Credit-card Fraud Detection System using Big Data Analytics," 2022 International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME), Maldives, Maldives, 2022, pp. 1-7, doi: 10.1109/ICECCME55909.2022.9987791.

[17] C. C. Meng, K. M. Lim, C. P. Lee and J. Y. Lim, "Credit Card Fraud Detection using TabNet," 2023 11th International Conference on Information and Communication Technology (ICoICT), Melaka, Malaysia, 2023, pp. 394-399, doi: 10.1109/ICoICT58202.2023.10262711.

[18] A. A. Taha and S. J. Malebary, "An Intelligent Approach to Credit Card Fraud Detection Using an Optimized Light Gradient Boosting Machine," in *IEEE Access*, vol. 8, pp. 25579-25587, 2020, doi: 10.1109/ACCESS.2020.2971354.

[19] 1.M. A. Alamri and M. A. Ykhlef, "A Machine Learning-Based Framework for Detecting Credit Card Anomalies and Fraud," 2023 27th International Conference on Information Technology (IT), Zabljak, Montenegro, 2023, pp. 1-7, doi: 10.1109/IT57431.2023.10078528.

[20] P. A. Jadhav, U. Lalwani, A. Gour, M. Shayan and S. Motwani, "Identification of Fraudulent Credit Card transactions using Machine Learning Algorithms," 2022 IEEE 7th International conference for Convergence in Technology (I2CT), Mumbai, ndia,2022, pp. 1-4, doi: 10.1109/I2CT54291.2022.9824165.

[21] Anderson M. (2008). UFrom Subprime Mortgages to Subprime Credit Cards‘. Communities and Banking, Federal Reserve Bank of Boston, pp. 21-23.

[22] Anwer et al. (2009-2010). UOnline Credit Card Fraud Prevention System for Developing Countries‘, International Journal of Reviews in Computing, ISSN: 2076-3328, Vol. 2, pp. 62-70.

[23] Arias, J.C. & Miller R. (2009). UMarket Analysis of Student about Credit Cards‘. Business Intelligence Journal, Vol. 3, No. 1, pp. 23-36.

[24] Bhatla T.P. et al. (2003). UUnderstanding Credit Card Frauds‘. Cards Business Review, 01, pp. 01-15.

[25] Bhusari V. & Patil S. (2011). UStudy of Hidden Markov Model in Credit Card Fraudulent Detection‘. International Journal of Computer Applications, Vol. 20, No. 5, pp. 33-36.

[26] Calem P. & Mester L. (1995). UConsumer Behavior and the Stickiness of Credit Card Interest Rates‘. The American Economic Review, Vol. 85, No. 5, pp. 1327-1333.

[27] Kodipalli, A., Guha, S., Dasar, S., & Ismail, T. (2022). An inception-ResNet deep learning approach to classify tumours in the ovary as benign and malignant. Expert Systems, e13215.

[28] Kodipalli, A., Devi, S., Dasar, S., & Ismail, T. (2022). Segmentation and classification of ovarian cancer based on conditional adversarial image to image translation approach. Expert Systems, e13193.

[29] Ruchitha, P. J., Sai, R. Y., Kodipalli, A., Martis, R. J., Dasar, S., & Ismail, T. (2022, October). Comparative analysis of active contour random walker and watershed algorithms in segmentation of ovarian cancer. In 2022 International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics (DISCOVER) (pp. 234-238). IEEE.

[30] Guha, S., Kodipalli, A., & Rao, T. (2022). Computational Deep Learning Models for Detection of COVID-19 Using Chest X-Ray Images. In Emerging Research in Computing, Information, Communication and Applications: Proceedings of ERCICA 2022 (pp. 291-306). Singapore: Springer Nature Singapore.